

## SAFETY DATA SHEET

Creation Date 16-April-2012

Revision Date 29-March-2024

Revision Number 3

### 1. Identification

**Product Name** 1-Bromo-3-methyl-2-butene

**Cat No. :** A13388

**CAS-No** 870-63-3  
**Synonyms** 3,3-Dimethylallyl bromide; Prenyl bromide

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Flammable liquids</b>                                | Category 3   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Flammable liquid and vapor

Causes severe skin burns and eye damage

May cause respiratory irritation



### Precautionary Statements

#### Prevention

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep container tightly closed

Ground/bond container and receiving equipment

Use explosion-proof electrical/ventilating/lighting/equipment

Do not breathe dust/fumes/gas/mist/vapours/spray

Wash face, hands and any exposed skin thoroughly after handling

Use only outdoors or in a well-ventilated area

Wear protective gloves/protective clothing/eye protection/face protection

Take action to prevent static discharges

#### Response

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Wash contaminated clothing before reuse

Fight fire with normal precautions from a reasonable distance

Evacuate area

#### Storage

Store locked up

Store in a well-ventilated place. Keep cool

Store in a closed container

#### Disposal

Dispose of contents/container to an approved waste disposal plant

### Other Hazards

Light sensitive

## 3. Composition/Information on Ingredients

| Component                   | CAS-No   | Weight % |
|-----------------------------|----------|----------|
| 2-Butene, 1-bromo-3-methyl- | 870-63-3 | >95      |

## 4. First-aid measures

### General Advice

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

### Inhalation

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the

substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Get medical attention.

**Most important symptoms/effects**

Causes burns by all exposure routes. Difficulty in breathing. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Notes to Physician**

Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Unsuitable Extinguishing Media**

No information available

**Flash Point**

32 °C / 89.6 °F

**Method -**

No information available

**Autoignition Temperature**

No information available

**Explosion Limits****Upper**

No data available

**Lower**

No data available

**Sensitivity to Mechanical Impact**

No information available

**Sensitivity to Static Discharge**

No information available

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen halides.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA**

**Health**  
3

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions**

Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## 7. Handling and storage

**Handling**

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not

breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Storage.**

Refrigerator/flammables. Corrosives area. Keep away from heat, sparks and flame. Protect from direct sunlight. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Butyl rubber   | recommendations   |                 |                        |
| Nitrile rubber |                   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

**Physical State**  
**Appearance**  
**Odor**

Liquid  
Dark yellow  
No information available

|                                        |                                          |
|----------------------------------------|------------------------------------------|
| Odor Threshold                         | No information available                 |
| pH                                     | No information available                 |
| Melting Point/Range                    | No data available                        |
| Boiling Point/Range                    | 82 - 83 °C / 179.6 - 181.4 °F @ 150 mmHg |
| Flash Point                            | 32 °C / 89.6 °F                          |
| Evaporation Rate                       | No information available                 |
| Flammability (solid,gas)               | Not applicable                           |
| Flammability or explosive limits       |                                          |
| Upper                                  | No data available                        |
| Lower                                  | No data available                        |
| Vapor Pressure                         | No information available                 |
| Vapor Density                          | 5.1                                      |
| Specific Gravity                       | 1.270                                    |
| Solubility                             | No information available                 |
| Partition coefficient; n-octanol/water | No data available                        |
| Autoignition Temperature               | No information available                 |
| Decomposition Temperature              | No information available                 |
| Viscosity                              | No information available                 |
| Molecular Formula                      | C5 H9 Br                                 |
| Molecular Weight                       | 149.04                                   |

## 10. Stability and reactivity

|                                  |                                                                                                                          |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                               |
| Stability                        | Stable under normal conditions. Light sensitive.                                                                         |
| Conditions to Avoid              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to light. |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Strong bases                                                                      |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen halides                                                |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                 |
| Hazardous Reactions              | None under normal processing.                                                                                            |

## 11. Toxicological information

### Acute Toxicity

|                       |                                                             |
|-----------------------|-------------------------------------------------------------|
| Product Information   | No acute toxicity information is available for this product |
| Component Information |                                                             |

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                      | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------|----------|------------|------------|------------|------------|------------|
| 2-Butene,<br>1-bromo-3-methyl- | 870-63-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                   |                          |
|-------------------|--------------------------|
| Mutagenic Effects | No information available |
|-------------------|--------------------------|

|                      |                           |
|----------------------|---------------------------|
| Reproductive Effects | No information available. |
|----------------------|---------------------------|

|                                                                  |                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Developmental Effects</b>                                     | No information available.                                                                                                                                                                                                                                                                                                                       |
| <b>Teratogenicity</b>                                            | No information available.                                                                                                                                                                                                                                                                                                                       |
| <b>STOT - single exposure</b><br><b>STOT - repeated exposure</b> | Respiratory system<br>None known                                                                                                                                                                                                                                                                                                                |
| <b>Aspiration hazard</b>                                         | No information available                                                                                                                                                                                                                                                                                                                        |
| <b>Symptoms / effects, both acute and delayed</b>                | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>                           | No information available                                                                                                                                                                                                                                                                                                                        |
| <b>Other Adverse Effects</b>                                     | The toxicological properties have not been fully investigated.                                                                                                                                                                                                                                                                                  |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                                                                 |
|--------------------------------------|-----------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely based on information available.         |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                       |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its volatility. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                                |                                                        |
|--------------------------------|--------------------------------------------------------|
| <b>UN-No</b>                   | UN2920                                                 |
| <b>Proper Shipping Name</b>    | consumer commodity Corrosive liquid, flammable, n.o.s. |
| <b>Technical Name</b>          | 2-Butene, 1-bromo-3-methyl-                            |
| <b>Hazard Class</b>            | 8                                                      |
| <b>Subsidiary Hazard Class</b> | 3                                                      |
| <b>Packing Group</b>           | II                                                     |

### TDG

|                                |                                     |
|--------------------------------|-------------------------------------|
| <b>UN-No</b>                   | UN2920                              |
| <b>Proper Shipping Name</b>    | Corrosive liquid, flammable, n.o.s. |
| <b>Hazard Class</b>            | 8                                   |
| <b>Subsidiary Hazard Class</b> | 3                                   |
| <b>Packing Group</b>           | II                                  |

### IATA

|                                |                                     |
|--------------------------------|-------------------------------------|
| <b>UN-No</b>                   | UN2920                              |
| <b>Proper Shipping Name</b>    | Corrosive liquid, flammable, n.o.s. |
| <b>Hazard Class</b>            | 8                                   |
| <b>Subsidiary Hazard Class</b> | 3                                   |
| <b>Packing Group</b>           | II                                  |

### IMDG/IMO

|                                |                                     |
|--------------------------------|-------------------------------------|
| <b>UN-No</b>                   | UN2920                              |
| <b>Proper Shipping Name</b>    | Corrosive liquid, flammable, n.o.s. |
| <b>Hazard Class</b>            | 8                                   |
| <b>Subsidiary Hazard Class</b> | 3                                   |
| <b>Packing Group</b>           | II                                  |

## 15. Regulatory information

### International Inventories

| Component                   | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 2-Butene, 1-bromo-3-methyl- | 870-63-3 | -   | X    | X    | ACTIVE                                        | 212-799-5 | -      | -   |

| Component                   | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------------------|----------|-------|----------|------|------|------|------|-------|-------|
| 2-Butene, 1-bromo-3-methyl- | 870-63-3 | -     | KE-03678 | X    | X    | X    | -    | X     | -     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

**Authorisation/Restrictions according to EU REACH**

Not applicable

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                   | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2-Butene, 1-bromo-3-methyl- | 870-63-3 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                   | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2-Butene, 1-bromo-3-methyl- | 870-63-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**